



IILM UNIVERSITY, GREATER NOIDA

INTERNSHIP REPORT PRESENTATION

Student Management System

Thiranex Internship Program | C++ Programming Track

P Vedansh

Roll No. 25SCS1003000705 | B.Tech CSE

25 July 2026 – 24 August 2026



P Vedansh

B.Tech – Computer Science
& Engineering

ABOUT THE STUDENT

Candidate Profile



Roll Number

25SCS1003000705



Institute

IILM University, Greater Noida, U.P.



Programme

B.Tech CSE, Batch 2025–2029



Internship Domain

C++ Programming (Thiranex)



Duration

25 July 2026 – 24 August 2026

Thiranex — Skill Development & Future Tech

Thiranex is an India-based education and technology platform that provides virtual internships, training programs, and project-based learning opportunities for students and aspiring technology professionals. Its programs give students practical exposure through project work, mentorship, and portfolio development. Thiranex operates as an MSME-registered company and describes its internship certificates as verifiable credentials, backed by a certificate-verification facility on its website.



MSME Registered

Operates as a registered MSME company under the Government of India.



Verifiable Certificates

Issues internship certificates with a QR-based verification facility.



Project-Based Learning

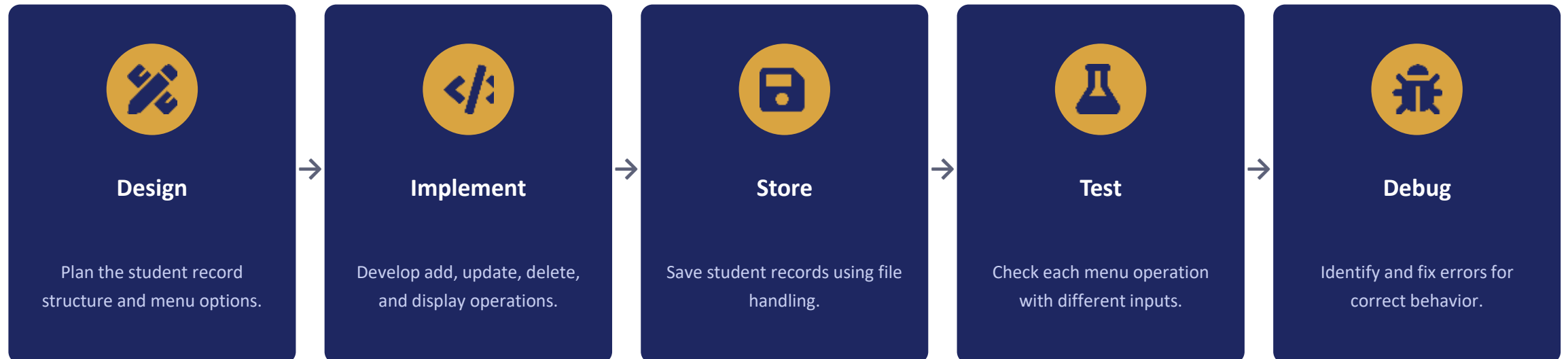
Focuses on hands-on projects, mentorship, and portfolio development.

Why a File-Based Student Manager



Managing student records manually becomes difficult as the number of students grows, especially when information needs to be added, updated, deleted, or viewed quickly. This project provides a simple C++-based system to organize student records and store them persistently using file handling instead of keeping everything only during program execution.

High-Level Workflow



What the Project Set Out to Achieve



Design Record Structure

Plan a clear structure for storing each student's information.



Menu-Driven Interface

Build an intuitive, structured console menu for navigation.



Implement CRUD Operations

Develop add, update, delete, and display functionality in C++.



Persistent File Storage

Store and retrieve records reliably using file handling.



Input Validation

Handle invalid or unexpected inputs gracefully across all menus.



Testing & Debugging

Verify every operation and resolve issues found during testing.

The Toolkit Behind the Project



C++

Core programming language



GCC

Compilation & execution



VS Code

Development environment



File I/O

Persistent data storage



ifstream / ofstream

Reading & writing files



Functions

Modular program structure



GitHub

Version control & hosting



Terminal

Running & testing the program

From Console Input to Stored Record



Input Layer

Console menu captures the student's choice



Validation Layer

Checks inputs before any operation runs



Operations Layer

Functions handle add, update, delete, display



File Storage Layer

ifstream / ofstream persist records to disk



Output Layer

Results are displayed back on the console

The Development Timeline

A phase-based approach carried out over the internship period

1

Design

Planned the student record structure and menu options.

2

Implement

Developed add, update, delete, and display operations in C++.

3

Store

Integrated file handling to save records persistently.

4

Test

Checked each menu operation with a range of inputs.

5

Debug

Identified and fixed errors to finalize the application.

Skills & Capabilities Gained



C++ File Handling & Persistent Storage



CRUD Operations on Student Records



Menu-Driven Application Development



Function-Based Program Structure



Input Validation & Error Handling



Testing & Debugging

CERTIFICATION

Successfully Completed



Track

C++ Programming



Certificate ID

THX-JUL2526-371



Duration

25 Jul 2026 – 24 Aug 2026





Thank You

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